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Cleartext at the *Edge*

A Neptune R900 case study in RF telemetry exposure

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RX 915.000 MHZ · RECEIVE ONLY



WHOAMI

Rick Ramirez

NOW	Lead product security engineer: OT/ICS, medical devices, cyber-physical systems
BEFORE	U.S. Marine Corps radio and electronics technician
RESEARCH	Passive RF telemetry, smart utility systems, cyber-physical privacy
CERTIFIED	GICSP: Global Industrial Cyber Security Professional

Scope and Boundaries

THIS TALK IS

- ✓ A passive RF case study
- ✓ A privacy and security analysis
- ✓ A look at systemic design risk
- ✓ A case for RF as attack surface

THIS TALK IS NOT

- – A guide to targeting meters
- – A release of anyone’s telemetry
- – An attempt to disrupt utilities
- – An exploitation walkthrough

THE PREMISE

The boundary is not the firewall. *The boundary is wherever the signal can be heard.*



02

CHAPTER 02 · RF

Primer

Why radio sits outside security visibility, and how a low-cost receiver makes it readable.



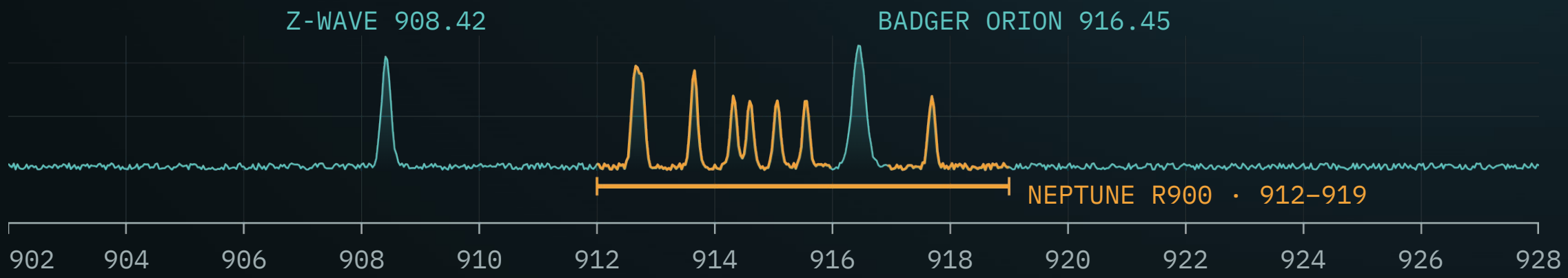
The Blind Spot

Security programs are built around the packet. RF happens before there is a packet.

If a system transmits, someone else may be able to receive it. None of your tools will record that they did.

CONTROL	NETWORK	RF BROADCAST
Firewall logs	Sees it	Never sees it
EDR	Sees it	Never sees it
SIEM	Sees it	Never sees it
Packet capture	Sees it	Never sees it
Asset inventory / NAC	Sees it	Never sees it

The 902–928 MHz Band



WHO SHARES IT

- Water meters
- Gas & electric meters
- SCADA radios
- Home automation
- Weather & soil sensors
- LoRaWAN nodes

$30–60

An RTL-SDR dongle, a wire antenna and open-source rtl_433 hear all of it.

03

CHAPTER 03 · NEPTUNE R900

Case Study

How cleartext water-meter telemetry surfaced in a garden sensor’s console.



How This Started

01

I was building a soil-moisture sensor for my garden, on 915 MHz.

02

While cleaning up its output, other packets appeared in the console.

03

They were water meters, including mine. In plaintext.

No targeting. No special tools. It simply showed up.



NEPTUNE MACH 10 · R900i
SERIAL MASKED

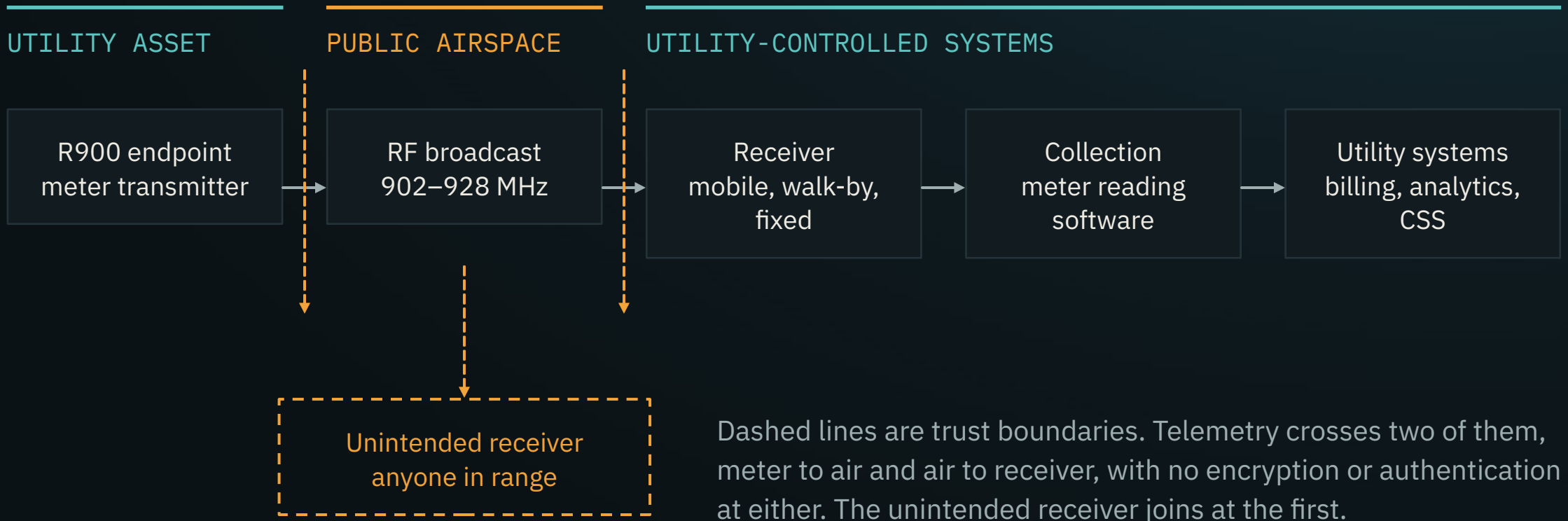
03 · CASE STUDY

Neptune R900

ROLE	Radio endpoint that reports a water meter’s reading
BAND	902–928 MHz ISM, frequency hopping
CADENCE	Mobile message about every 14 s; fixed-network message every 7.5 min
IDENTITY	Static ID tied to the meter’s serial number
PROTECTION	No encryption. No message authentication.
LIFESPAN	10–20 years in service

Sources: Neptune product literature; City of Oneida, NY meter memo, Feb 2026

AMR Data Flow and Trust Boundaries



Dashed lines are trust boundaries. Telemetry crosses two of them, meter to air and air to receiver, with no encryption or authentication at either. The unintended receiver joins at the first.

What a Decoded Frame Can Reveal

EXAMPLE JSON OUTPUT · FIELD NAMES ONLY

```
{
  "model": "Neptune-R900",
  "id": "Unique Meter ID",
  "consumption": "Water Usage Amount",
  "leak": 0,
  "backflow": 0,
  "rssi": "Signal Strength",
  "time": "Time signal was transmitted"
}
```

IDENTITY

A meter ID, static and tied to the serial number

USAGE

A consumption reading, repeated over time

STATUS

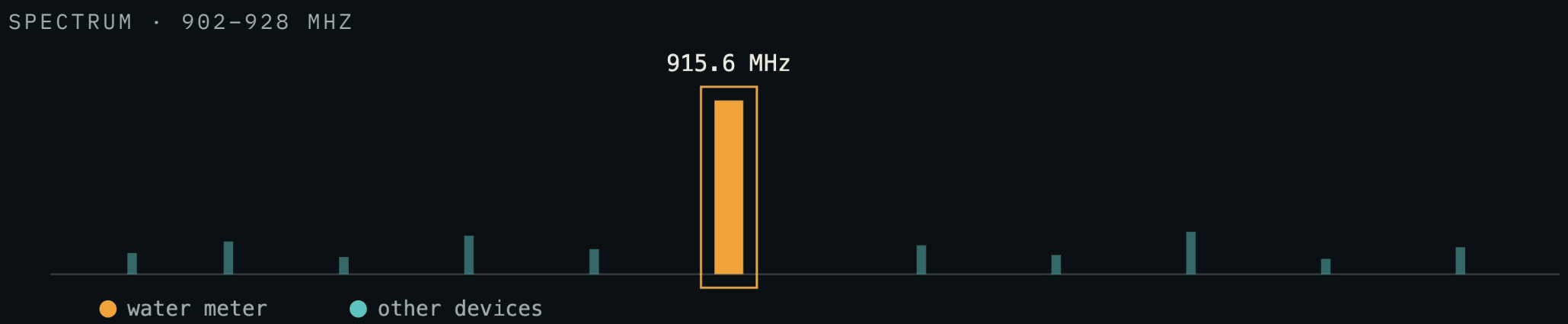
Leak and backflow or tamper-style flags, where present

SIGNAL

Signal strength and a timestamp

From Radio Wave to Readable Data

A water meter sends a short radio burst.



1 SIGNAL
hear a burst

2 DEMODULATE
wave → symbols

3 BITS
symbols → 1s, 0s

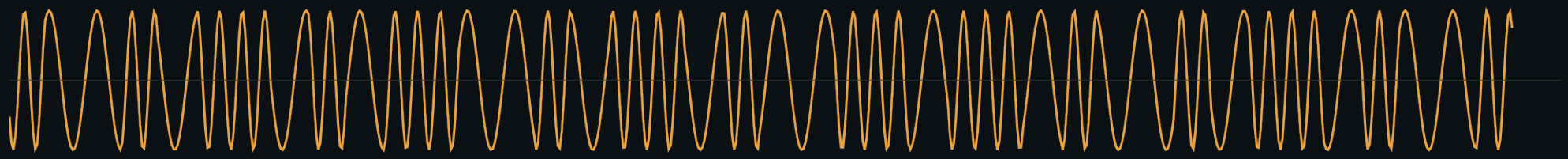
4 DECODE
find the frame

5 FIELDS
readable values

From Radio Wave to Readable Data

The receiver turns the wave into symbols.

Frequency shifts carry the symbols



Nothing is decrypted — the wave simply **is** the data

1 SIGNAL
hear a burst

2 DEMODULATE
wave → symbols

3 BITS
symbols → 1s, 0s

4 DECODE
find the frame

5 FIELDS
readable values

From Radio Wave to Readable Data

Symbols become a raw stream of 1s and 0s.

The symbols become a raw bitstream

```
1 0 0 1 0 1 1 0 1 0 1 1 0 0 1 0 1 1 0 1 0 0 1 1
0 1 1 0 1 0 0 1 0 1 1 0 1 0 0 1 0 1 1 0 1 0 0 1
0 0 1 0 1 1 0 1 0 0 1 0 1 1 0 1 0 0 1 0 1 1 0 1
```

96 bits, in the clear

1 SIGNAL hear a burst → **2 DEMODULATE** wave → symbols → **3 BITS** symbols → 1s, 0s → **4 DECODE** find the frame → **5 FIELDS** readable values

From Radio Wave to Readable Data

The decoder finds the frame and verifies it.

The decoder finds the frame and checks it

```
1 0 0 1 0 1 1 0 1 0 1 1 0 0 1 0 1 1 0 1 0 0 1 1
0 1 1 0 1 0 0 1 0 1 1 0 1 0 0 1 0 1 1 0 1 0 0 1
0 0 1 0 1 1 0 1 0 0 1 0 1 1 0 1 0 0 1 0 1 1 0 1
```

▸ frame start found checksum OK ✓ no auth · no encryption

1 SIGNAL
hear a burst

2 DEMODULATE
wave → symbols

3 BITS
symbols → 1s, 0s

4 DECODE
find the frame

5 FIELDS
readable values

From Radio Wave to Readable Data

Named fields appear — no key or password needed.

DECODED FRAME · SIMULATED, SANITIZED

Model	Neptune-R900	
Meter ID	••••4821	never changes → trackable
Water reading	0043531	rises over time → routine
Leak flag	0	
Backflow flag	0	
Time	14:02:37	

1 SIGNAL hear a burst → **2 DEMODULATE** wave → symbols → **3 BITS** symbols → 1s, 0s → **4 DECODE** find the frame → **5 FIELDS** readable values

Everyone in Range Hears It

One meter sends its reading — a short burst on 915 MHz, about every 14 seconds.



Everyone in Range Hears It

The utility collector on the pole receives it. That is the intended path.



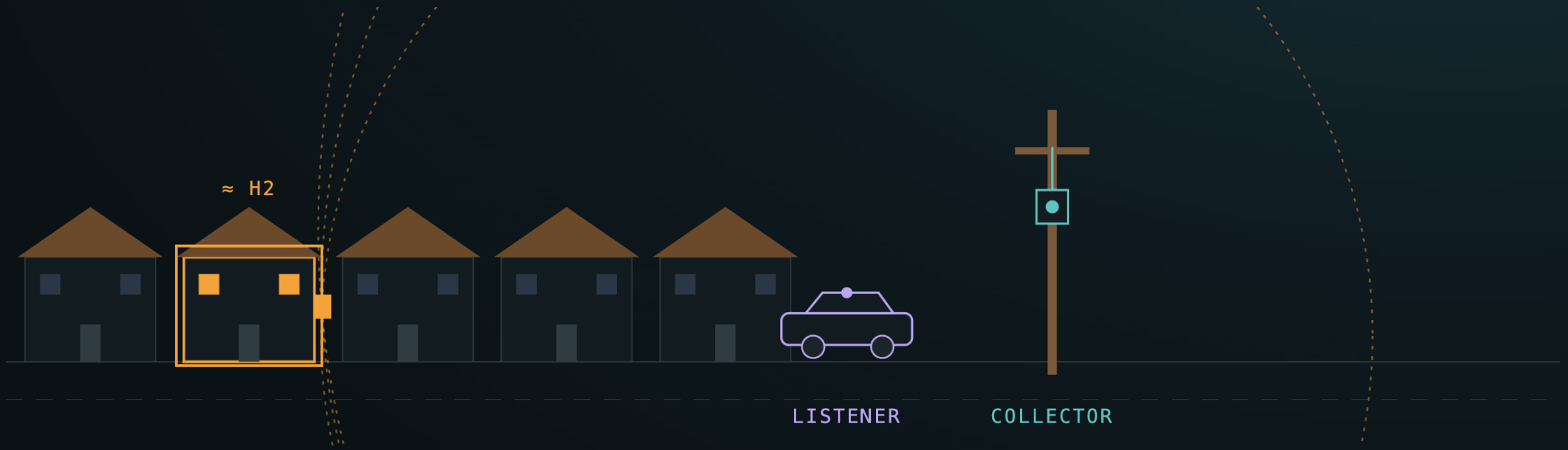
Everyone in Range Hears It

But the same packet reaches anyone in range. The meter can't tell who is listening.



Everyone in Range Hears It

With a few listening points, signal strength narrows the burst to one house.



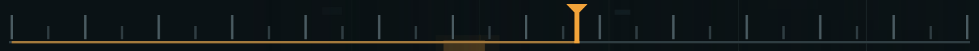
Anonymous reading + permanent ID + signal strength = which house.

04

CHAPTER 04 · PATTERN OF LIFE

Risk

What the telemetry reveals, and how the exposure rates.



Telemetry Can Become Pattern-of-Life Data





DAILY ROUTINES

Usage can reveal occupancy patterns and daily life

INDUSTRIAL SITES

Consumption can hint at manufacturing, treatment, or other sensitive processes

VACANT AND AWAY

Usage patterns show when a building is likely empty

Potential inferences from repeated consumption data

Threat Model: The RF Segment

STRIDE	AT THE AIR BOUNDARY	STATUS	RISKS
Info disclosure	Cleartext reading, static ID, status flags	Observed, passive	1–6
Spoofing	No message authentication on frames	Not tested	7
Tampering	No replay or integrity protection	Not tested	7
Repudiation	No proof a frame came from the meter	Inferred	7
Denial of service	Jamming or junk frames block real reads	Not tested	—
Elevation of privilege	One-way broadcast, no command path	Not applicable	—

Only disclosure was observed. Everything else comes from the protocol design, not from testing.

CVSS vs. Context

V1 · CLEARTEXT · RISKS 1–6

8.3 High

CVSS:3.1/AV:A/AC:L/PR:N/UI:N
S:U/C:H/I:N/A:N · CR:H

V2 · NO AUTH · RISK 7

5.0 Medium

CVSS:3.1/AV:A/AC:L/PR:N/UI:N
S:U/C:N/I:L/A:N · IR:H

#	RISK	CVSS	CONTEXT	RATING
1	Passive surveillance	8.3	20 5×4	High
2	Static identifiers	8.3	20 5×4	High
3	Cleartext telemetry	8.3	20 5×4	High
4	Pattern of life	8.3	16 4×4	High
5	Asset enumeration	8.3	12 4×3	Medium–High
6	Weak trust boundaries	8.3	16 4×4	High
7	Integrity concerns	5.0	12 3×4	Medium–High

CVSS 3.1 environmental, utility context · Context = L × I, 1–5 each

Context lifts cleartext to High.

Transmit Risks: Possible, Not Tested

NOT TESTED

Every row needs a transmitter. This research never transmitted, so none of these is demonstrated.

V2 · NO AUTH · RISK 7

ONE METER

4.3–6.0 Medium

TOWN FLEET · 10K+ METERS

6.5–8.5 Up to High

CVSS:3.1/AV:A/AC:L/PR:N/UI:N/S:U · IR:H

#	RISK	METER	FLEET	RATING
T1	Spoofed readings	5.0	8.3	Medium–High
T2	Replayed frames	5.0	8.3	Medium–High
T3	Forged leak or tamper flags	5.0	8.3	Medium–High
T4	Suppressed backflow alerts	5.0	8.3	High
T5	Collector flooding	6.0	8.5	Medium
T6	Read denial (jamming)	4.3	6.5	Low–Medium

Meter: one device · Fleet: town deployment, I:H or A:H

Medium per meter. High across a town.

05

CHAPTER 05 · BEYOND ONE METER

Implications

Why it matters, what was disclosed, and what security teams should do.

Disclosure Scope

FEB 20, 2026

Cleartext R900 packets observed while testing a garden sensor

FEB 2026

CERT/CC coordinated disclosure opened, VU#318962

DURING PROCESS

Outreach to Neptune Technology Group: no response

MAY 2026

Closed as “functions as designed.” No CVE, no advisory.

BROADER DISCLOSURE · AMR/AMI VENDORS

Itron

Water, gas, electric

Neptune

Water, sewer. Primary case study

Badger Meter

Water, sewer

Landis+Gyr

Electric. Sent a Customer Information Letter to impacted customers

05 · IMPLICATIONS

Still Being Installed in 2026

FEB 2026

Oneida, NY

Standardizes on MACH 10 R900i across 4,600 connections

MAY 2026

CERT/CC closes the case

“Functions as designed.” No CVE.

JUN 2026

Enid, OK

Sole-source, up to $600K, to stay compatible with Neptune AMI

SEP 2026

Island Lake, IL

Approves an R900 gateway for its Neptune upgrade

UNDERWAY

Colona, IL

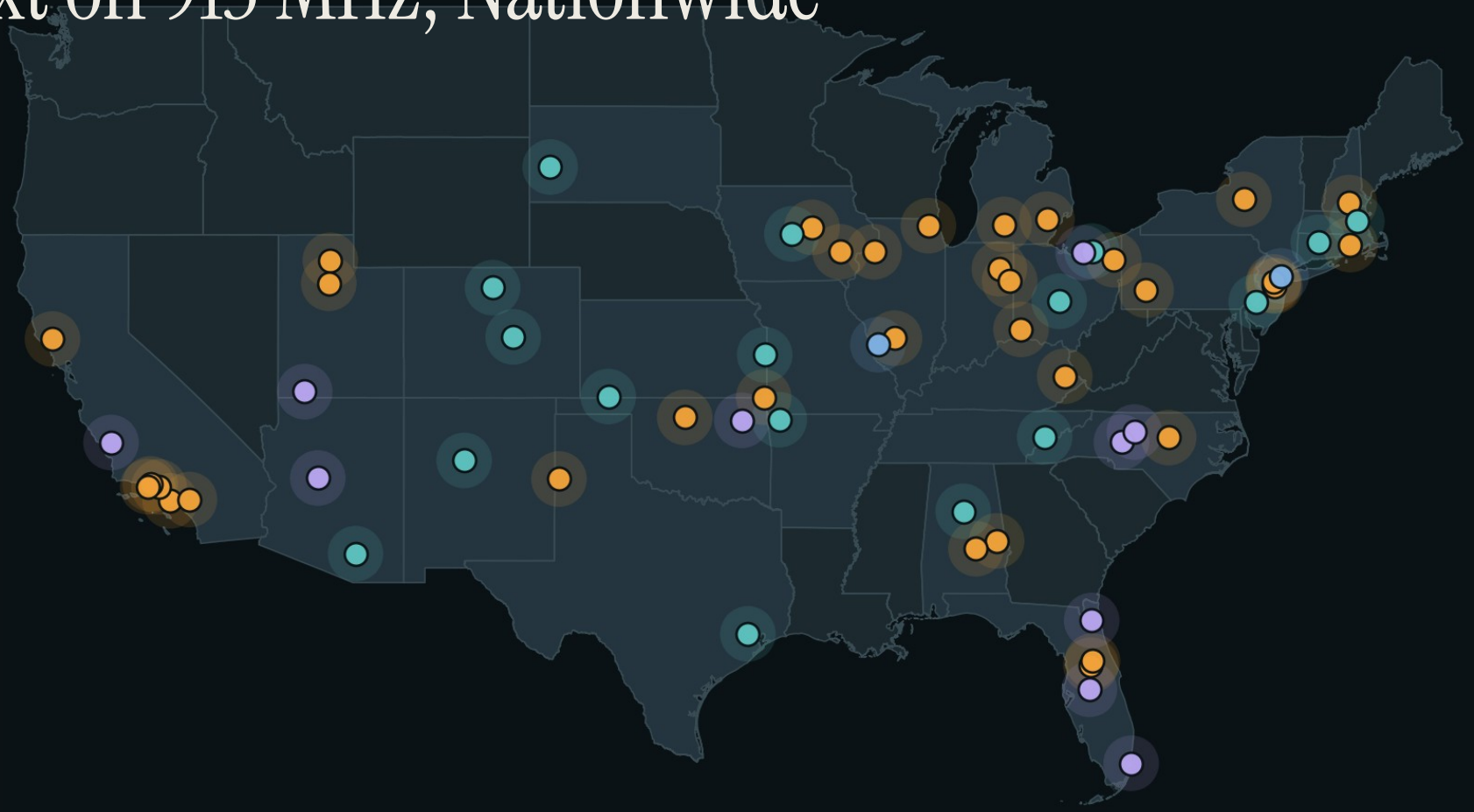
Citywide E-Coder R900i rollout, about 2,100 meters

Closed as designed. Still being bought.

Sources: city council agendas, purchase memos and program pages, 2026. A sample of public records, not a full count. None of the documents reviewed set encryption or authentication requirements.

05 · IMPLICATIONS · THE WHOLE BAND

Cleartext on 915 MHz, Nationwide



Neptune R900 · Itron ERT · Badger ORION · Landis+Gyr — + consumer IoT on the same band: weather · soil · TPMS

Shaded = confirmed-deployment state · 64 deployments · 28 states · a sample, not a census CH 30 · 915.00 MHZ

A Free Tool Already Decodes It

~380

device decoders ship with rtl_433, the free, open-source receiver

4 / 4

meter families on the map have a built-in decoder — no reverse engineering

1

protocol in that list is flagged encrypted — a single car keyfob

BUILT-IN DECODERS · 902–928 MHz

Itron ERT [149/154/160/161] Neptune R900 [228/385] Badger ORION [223/282/290] Landis+Gyr [271–273]

+ hundreds of consumer sensors on the band — Acurite, LaCrosse, TPMS, and the Ecowitt soil family that started this research

The decoders are already in the box.

Decoder counts and protocol IDs from the public rtl_433 project · one open-source tool, a $25 SDR

The Scale of the Exposure

80M+

Endpoints on the 915 MHz band across the four families: Neptune, Itron, Badger, Landis+Gyr

10–20

Years each radio stays in the field, so today's design is locked in for a generation

0

Purchase records reviewed that required encryption or authentication

One band. Tens of millions of endpoints. No key.

Endpoint total is the sum of the four families' estimates from vendor literature and the disclosure research

What Security Teams Should Take Away

PROGRAM & PROCUREMENT

01 Treat RF telemetry as an external-facing interface

02 Include RF in threat modeling and architecture reviews

03 Add RF security requirements to procurement language

04 Run privacy impact assessments for AMR/AMI deployments

05 Monitor for abnormal RF behavior where feasible

DESIGN CONTROLS

- ✓ Encrypt sensitive telemetry payloads
- ✓ Authenticate messages
- ✓ Add replay protection
- ✓ Avoid static identifiers where possible
- ✓ Reduce unnecessary broadcast fields

So What Can You Do?

IF YOU'RE A RESIDENT

- Ask your utility: is meter telemetry encrypted? Is an opt-out or non-broadcasting meter offered?
- Raise it at utility board or PUC meetings — procurement is public, and ratepayers have standing
- Keep the risk in proportion: this is privacy, not someone shutting off your water

IF YOU'RE IN SECURITY

- Add the signal layer to your threat models and architecture reviews
- Research receive-only and know the legal line; coordinate disclosure, don't transmit
- Push RF security into any procurement language you can influence

IF YOU'RE A UTILITY

- Require encryption, message authentication and replay protection at purchase
- Prefer rotating or pseudonymous IDs; cut unnecessary broadcast fields
- Plan migration now — these radios stay in the field 10 to 20 years

THE TAKEAWAY

RF is part of *the attack surface.*

Security reviews should include the signal layer.

QUESTIONS & DISCUSSION · END OF TRANSMISSION

Thank *you.*

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